

Q1 - 25 July - Shift 1

The volume charge density of a sphere of radius 6 m is $2 \mu\text{C cm}^{-3}$. The number of lines of force per unit surface area coming out from the surface of the sphere is $\underline{\hspace{2cm}} \times 10^{10} \text{ NC}^{-1}$.

[Given : Permittivity of vacuum

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}]$$

Space for your notes:

Q2 - 26 July - Shift 2

Two uniformly charged spherical conductors A and B of radii 5 mm and 10 mm are separated by a distance of 2 cm. If the spheres are connected by a conducting wire, then in equilibrium condition, the ratio of the magnitudes of the electric fields at the surface of the sphere A and B will be :

(A) 1 : 2 (B) 2 : 1 (C) 1 : 1 (D) 1 : 4

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Q3 - 26 July - Shift 2

Three point charges of magnitude $5\mu\text{C}$, $0.16\mu\text{C}$ and $0.3\mu\text{C}$ are located at the vertices A, B, C of a right angled triangle whose sides are $AB = 3\text{cm}$, $BC = 3\sqrt{2}\text{ cm}$ and $CA = 3\text{ cm}$ and point A is the right angle corner. Charge at point A experiences $\underline{\hspace{2cm}}$ N of electrostatic force due to the other two charges.

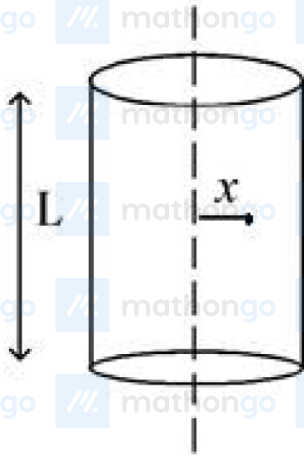
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Questions

MathonGo

Q4 - 27 July - Shift 1

A long cylindrical volume contains a uniformly distributed charge of density $\rho \text{ Cm}^{-3}$. The electric field inside the cylindrical volume at a distance $x = \frac{2\epsilon_0}{\rho}$ m from its axis is _____ Vm^{-1}

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Q5 - 27 July - Shift 2

A charge of $4 \mu\text{C}$ is to be divided into two. The distance between the two divided charges is constant. The magnitude of the divided charges so that the force between them is maximum, will be:

- (A) $1 \mu\text{C}$ and $3 \mu\text{C}$ (B) $2 \mu\text{C}$ and $2 \mu\text{C}$
(C) 0 and $4 \mu\text{C}$ (D) $1.5 \mu\text{C}$ and $2.5 \mu\text{C}$

Space for your notes:

Q6 - 28 July - Shift 1

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Two electric dipoles of dipole moments 1.2×10^{-30} cm and 2.4×10^{-30} cm are placed in two different uniform electric fields of strengths 5×10^4 NC⁻¹ and 15×10^4 NC⁻¹ respectively. The ratio of maximum torque experienced by the electric dipoles will be $\frac{1}{x}$. The value of x is

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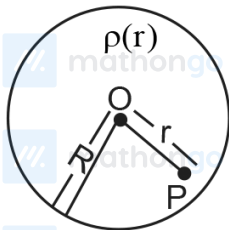
Q7 - 29 July - Shift 1

A spherically symmetric charge distribution is considered with charge density varying as

$$\rho(r) = \begin{cases} \rho_0 \left(\frac{3}{4} - \frac{r}{R} \right) & \text{for } r \leq R \\ \text{Zero} & \text{for } r > R \end{cases}$$

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Where, r ($r < R$) is the distance from the centre O (as shown in figure). The electric field at point P will be :



- (A) $\frac{\rho_0 r}{4\epsilon_0} \left(\frac{3}{4} - \frac{r}{R} \right)$ (B) $\frac{\rho_0 r}{3\epsilon_0} \left(\frac{3}{4} - \frac{r}{R} \right)$
 (C) $\frac{\rho_0 r}{4\epsilon_0} \left(1 - \frac{r}{R} \right)$ (D) $\frac{\rho_0 r}{5\epsilon_0} \left(1 - \frac{r}{R} \right)$

Q8 - 29 July - Shift 1

Given below are two statements.

Statement I : Electric potential is constant within and at the surface of each conductor.

Statement II : Electric field just outside a charged conductor is perpendicular to the surface of the conductor at every point.

In the light of the above statements, choose the most appropriate answer from the options give below.

- (A) Both statement I and statement II are correct
- (B) Both statement I and statement II are incorrect
- (C) Statement I is correct but statement II is incorrect
- (D) Statement I is incorrect but and statement II is correct

Official Ans. by NTA (A)

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Q9 - 29 July - Shift 2

Two identical metallic spheres A and B when placed at certain distance in air repel each other with a force of F . Another identical uncharged sphere C is first placed in contact with A and then in contact with B and finally placed at midpoint between spheres A and B. The force experienced by sphere C will be :

- (A) $3F/2$
- (B) $3F/4$
- (C) F
- (D) $2F$

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Answer Key

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Q1 (45) **Q2 (B)** **Q3 (17)** **Q4 (1)**
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Q5 (B) **Q6 (6)** **Q7 (C)** **Q8 (A)**
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Q9 (B)
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Q1 (45)

No. of electric field lines per unit area = electric field.

$$E = \frac{\rho r}{3\epsilon_0}, \text{ for } r = R$$

$$E = \frac{\rho R}{3\epsilon_0} = \frac{2 \times 6}{3 \times 8.85 \times 10^{-12}} = 0.45 \times 10^{12} \text{ NC}^{-1}$$

$$= 45 \times 10^{10} \text{ N/C}$$

Q2 (B)

$$V_A = V_B$$

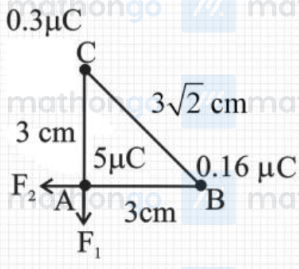
$$\frac{KQ_A}{R_A} = \frac{KQ_B}{R_B}$$

$$\frac{Q_A}{Q_B} = \frac{R_A}{R_B} = \frac{1}{2}$$

$$E_A = \frac{KQ_A}{R_A^2}; E_B = \frac{KQ_B}{R_B^2}$$

$$\frac{E_A}{E_B} = \frac{Q_A}{Q_B} \times \frac{R_B^2}{R_A^2} = \frac{R_B}{R_A} = \frac{2}{1}$$

Q3 (17)



$$F_1 = \frac{k \times 5 \times 0.3 \times 10^{-12}}{9 \times 10^{-4}}$$

$$= \frac{9 \times 10^9 \times 5 \times 0.3 \times 10^{-12}}{9 \times 10^{-4}}$$

$$= 1.5 \times 10 = 15 \text{ N}$$

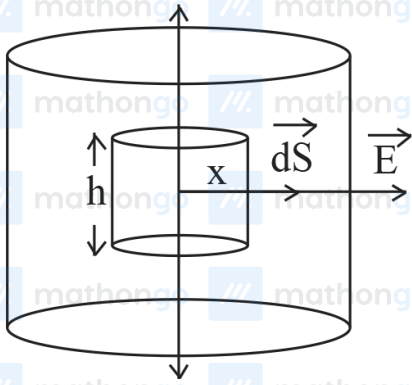
$$F_2 = \frac{9 \times 10^9 \times 5 \times 0.16 \times 10^{-12}}{9 \times 10^{-4}} = 8 \text{ N}$$

force experienced by charge at A = $\sqrt{F_1^2 + F_2^2}$

$$= \sqrt{15^2 + 8^2}$$

$$= \sqrt{289} = 17 \text{ N}$$

Q4 (1)



$$\int E ds \cos 0 = \frac{q}{\epsilon_0}$$

$$\Rightarrow E \cdot 2\pi x h = \frac{\rho \times \pi x^2 h}{\epsilon_0}$$

$$\Rightarrow E = \frac{\rho x}{2\epsilon_0}$$

$$\Rightarrow E = \frac{\rho}{2\epsilon_0} \times \frac{2\epsilon_0}{\rho} = 1$$

Q5 (B)



$$F = \frac{Kq(4-q)}{d^2}$$

$$\frac{dF}{dq} = \frac{K}{d^2} [4 - 2q] = 0$$

$$q = 2$$

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Q6 (6)

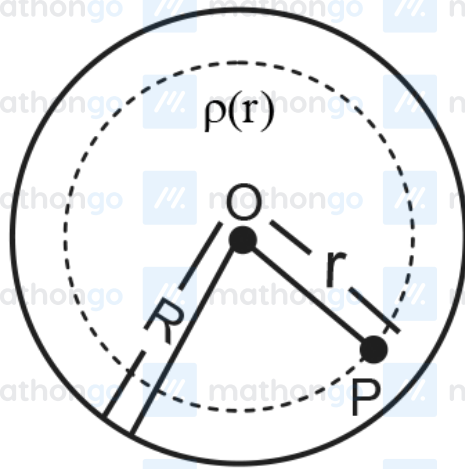
$$|\tau|_{\max} = PE$$

$$\frac{\tau_1}{\tau_2} = \frac{P_1 E_1}{P_2 E_2} = \frac{1 \cdot 2 \times 10^{-30} \times 5 \times 10^4}{2 \cdot 4 \times 10^{-30} \times 15 \times 10^4} = \frac{1}{6}$$

Hence $x = 6$

Q7 (C)

By Gauss law



$$\oint \vec{E} \cdot d\vec{s} = \frac{Q_{in}}{\epsilon_0}$$

$$E \cdot 4\pi r^2 = \frac{\int_0^r \rho_o \left(\frac{3}{4} - \frac{r}{R} \right) 4\pi r^2 dr}{\epsilon_0}$$

$$E 4\pi r^2 = \frac{\rho_o 4\pi}{\epsilon_0} \left(\frac{3}{4} \frac{r^3}{3} - \frac{r^4}{4R} \right)$$

$$E r^2 = \frac{\rho_o r^3}{4\epsilon_0} \left\{ 1 - \frac{r}{R} \right\}$$

$$E = \frac{\rho_o r}{4\epsilon_0} \left\{ 1 - \frac{r}{R} \right\}$$

Q8 (A)

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(Properties of conductor)

Statement – 1, true as body of conductor acts as equipotential surface.

Statement – 2 True, as conductor is equipotential.

Tangential component of electric field should be zero. Therefore electric field should be perpendicular to surface.

Q9 (B)

Let $q_A = q_B = q$



$$F = \frac{Kq^2}{r^2}$$

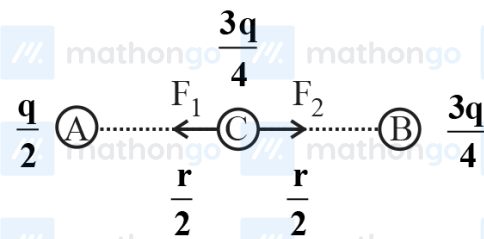
When C is placed in contact with A, charge on A

& C will be = $\frac{q}{2}$

Now C is placed in contact with B, charge on B &

C will be = $\frac{q + \frac{q}{2}}{2} = \frac{3q}{4}$

Now,



$$F' = F_2 - F_1 = \frac{\left(K \frac{3q}{4} - K \frac{q}{2} \right)}{\frac{r^2}{4}} \cdot \frac{3q}{4}$$

$$= \frac{3Kq^2}{4r^2} = \frac{3F}{4} \quad (\text{B})$$

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